

Module Five Journal

Natalie English

CS-499: Computer Science Capstone

Mr. Jeff Phillips

Southern New Hampshire University

02 August 2026

What is the significance of quantum computing?

Quantum computing holds significance as it addresses intricate issues at an exponentially quicker pace than traditional computers by employing subatomic physics, particularly superposition, entanglement, and qubits. Principal areas of influence comprise replicating precise molecular and chemical interactions to develop novel medications and vaccines within days rather than decades, overseeing extensive logistics, traffic management, and financial risk assessments that exceed the capabilities of conventional supercomputers, and establishing highly secure communication while necessitating a total overhaul of worldwide data encryption.

How will quantum computing change the field of computer science?

Quantum computing will radically alter computer science by reshaping cybersecurity, revolutionizing algorithm development, and transitioning computational models to quantum-classical hybrid systems. Issues that cannot be solved efficiently on traditional computers might be addressed in polynomial (efficient) time through quantum parallelism, superposition, and entanglement (see Grover's search algorithm and the P vs NP problem). This indicates that quantum processors may complement classical systems to manage demanding optimization and simulation tasks (effectively lowering the overall workload for both people and computers), but they could also pose a risk to commonly employed public-key cryptography, necessitating significant adjustments to cybersecurity protocols.

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Why is quantum computing relevant to your career interests or aspirations?

Quantum computing will impact my aspired career in web development by revolutionizing fundamental concepts such as cybersecurity, speeding up backend data processing, and enhancing artificial intelligence systems. Although web browsers won't operate directly on quantum hardware, cloud-based web applications and servers will use quantum or quantum-hybrid systems for intensive computational tasks.

Artifact Status Checkpoints

Checkpoint	Software Design and Engineering	Algorithms and Data Structures	Databases
Name of Artifact Used	Travl'r Getaways	Travl'r Getaways	Travl'r Getaways
Status of Initial Enhancement	Complete; finished debugging and polishing code.	Complete; implemented the basic framework needed to interact with the app's backend.	In progress; database interaction yet to be implemented.
Status of Final Enhancement	In Progress; database interaction yet to be implemented, and quality assurance testing needed.	In Progress	In Progress
Uploaded to ePortfolio	Yes, as a beta pre-release.	Yes, as a beta pre-release.	Yes, as a beta pre-release.

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Status of Finalized ePortfolio	In Progress	In Progress	In Progress
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References

Milvus. (2026). *What is Grover's algorithm, and what is its purpose?* Milvus.
<https://milvus.io/ai-quick-reference/what-is-grovers-algorithm-and-what-is-its-purpose>

Poddar, H. (2022, May 27). *Polynomial Vs Non-Polynomial Problem*. Medium.
<https://medium.com/@harsh.poddar201/p-vs-np-problem-3fb0031e03d3>

Schneider, J., & Smalley, I. (2025, June 10). *What is quantum computing?*. IBM .
<https://www.ibm.com/think/topics/quantum-computing>